

MODULE-I: FIRST ORDER ORDINARY DIFFERENTIAL EQUATIONS (09)

Exact differential equations, equations reducible to exact differential equations, linear and Bernoulli's equations, orthogonal trajectories (only in cartesian coordinates). Applications: Newton's law of cooling, law of natural growth and decay.

MODULE-II: ORDINARY DIFFERENTIAL EQUATIONS OF HIGHER ORDER (10)

Higher order linear differential equations with constant coefficients: non-homogeneous terms of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax}V(x)$ and $x V(x)$, method of variation of parameters.

MODULE-III: LAPLACE TRANSFORMS (10)

Laplace transforms: Laplace transform of standard functions, first shifting theorem, Laplace transforms of functions multiplied by 't' and divided by 't', Laplace transforms of derivatives and integrals of function, evaluation of integrals by Laplace transforms, Laplace transform of periodic functions.

Inverse Laplace transform by different methods, Convolution theorem (without proof). Applications: solving initial value problems by Laplace transform method

MODULE-IV: VECTOR DIFFERENTIATION (09)

Vector point functions and scalar point functions, gradient, divergence and curl, directional derivatives, vector identities, scalar potential functions, solenoidal and irrotational vectors.

MODULE-V: VECTOR INTEGRATION (10)

Line, surface and volume integrals. theorems of Green, Gauss and Stokes (without proofs) and their applications.